

- 4.46** (a) Silicon at $T = 300$ K is uniformly doped with boron atoms to a concentration of $3 \times 10^{16} \text{ cm}^{-3}$ and with arsenic atoms to a concentration of $1.5 \times 10^{16} \text{ cm}^{-3}$. Is the material n type or p type? Calculate the thermal equilibrium concentrations of majority and minority carriers. (b) Additional impurity atoms are added such that holes are the majority carrier and the thermal equilibrium concentration is $p_0 = 5 \times 10^{16} \text{ cm}^{-3}$. What type and concentration of impurity atoms must be added? What is the new value of n_0 ?

(a) $N_a > N_d \Rightarrow \text{p-type}$

Majority carriers are holes

$$p_o = N_a - N_d = 3 \times 10^{16} - 1.5 \times 10^{16} = 1.5 \times 10^{16} \text{ cm}^{-3}$$

Minority carriers are electrons

$$n_o = \frac{n_i^2}{p_o} = \frac{(1.5 \times 10^{10})^2}{1.5 \times 10^{16}} = 1.5 \times 10^4 \text{ cm}^{-3}$$

(b) Boron atoms must be added

$$p_o = N'_a + N_a - N_d \quad 5 \times 10^{16} = N'_a + 3 \times 10^{16} - 1.5 \times 10^{16}$$

$$\text{So } N'_a = 3.5 \times 10^{16} \text{ cm}^{-3} \quad n_o = \frac{(1.5 \times 10^{10})^2}{5 \times 10^{16}} = 4.5 \times 10^3 \text{ cm}^{-3}$$

- 4.55** (a) Silicon at $T = 300$ K is doped with donor impurity atoms at a concentration of $N_d = 6 \times 10^{15} \text{ cm}^{-3}$. (i) Determine $E_c - E_F$. (ii) Calculate the concentration of additional donor impurity atoms that must be added to move the Fermi energy level a distance kT closer to the conduction band edge. (b) Repeat part (a) for GaAs if the original donor impurity concentration is $N_d = 1 \times 10^{15} \text{ cm}^{-3}$.

(a) Silicon

$$(i) E_c - E_F = kT \ln \left(\frac{N_c}{N_d} \right) = (0.0259) \ln \left(\frac{2.8 \times 10^{19}}{6 \times 10^{15}} \right) = 0.2188 \text{ eV}$$

$$(ii) E_c - E_F = 0.2188 - 0.0259 = 0.1929 \text{ eV}$$

$$N_d = N_c \exp \left[\frac{-(E_c - E_F)}{kT} \right] = (2.8 \times 10^{19}) \exp \left[\frac{-0.1929}{0.0259} \right] \quad N_d = 1.631 \times 10^{16} \text{ cm}^{-3} = N'_d + 6 \times 10^{15}$$

$$\Rightarrow N'_d = 1.031 \times 10^{16} \text{ cm}^{-3} \quad \text{Additional donor atoms}$$

(b) GaAs

$$(i) E_c - E_F = (0.0259) \ln \left(\frac{4.7 \times 10^{17}}{10^{15}} \right) = 0.15936 \text{ eV}$$

$$(ii) E_c - E_F = 0.15936 - 0.0259 = 0.13346 \text{ eV}$$

$$N_d = (4.7 \times 10^{17}) \exp\left[\frac{-0.13346}{0.0259}\right] = 2.718 \times 10^{15} \text{ cm}^{-3} = N'_d + 10^{15}$$

$$\Rightarrow N'_d = 1.718 \times 10^{15} \text{ cm}^{-3} \text{ Additional donor atoms}$$

5.23 Consider three samples of silicon at $T = 300 \text{ K}$. The n-type sample is doped with arsenic atoms to a concentration of $N_d = 5 \times 10^{16} \text{ cm}^{-3}$. The p-type sample is doped with boron atoms to a concentration of $N_a = 2 \times 10^{16} \text{ cm}^{-3}$. The compensated sample is doped with both the donors and acceptors described in the n-type and p-type samples. (a) Find the equilibrium electron and hole concentrations in each sample, (b) determine the majority carrier mobility in each sample, (c) calculate the conductivity of each sample, (d) and determine the electric field required in each sample to induce a drift current density of $J = 120 \text{ A/cm}^2$.

$$\text{n-type: } \mu_n \cong 1100 \text{ cm}^2/\text{V-s} \quad \text{p-type: } \mu_p \cong 400 \text{ cm}^2/\text{V-s} \quad \text{compensated: } \mu_n \cong 1000 \text{ cm}^2/\text{V-s}$$

$$\text{(a) n-type: } n_o = N_d = 5 \times 10^{16} \text{ cm}^{-3}$$

$$p_o = \frac{n_i^2}{n_o} = \frac{(1.5 \times 10^{10})^2}{5 \times 10^{16}} = 4.5 \times 10^3 \text{ cm}^{-3}$$

$$\text{p-type: } p_o = N_a = 2 \times 10^{16} \text{ cm}^{-3} \quad n_o = \frac{(1.5 \times 10^{10})^2}{2 \times 10^{16}} = 1.125 \times 10^4 \text{ cm}^{-3}$$

$$\text{compensated: } n_o = N_d - N_a = 5 \times 10^{16} - 2 \times 10^{16} = 3 \times 10^{16} \text{ cm}^{-3}$$

$$p_o = \frac{(1.5 \times 10^{10})^2}{3 \times 10^{16}} = 7.5 \times 10^3 \text{ cm}^{-3}$$

$$\text{(b) n-type: } \sigma = e\mu_n n_o = (1.6 \times 10^{-19})(1100)(5 \times 10^{16}) = 8.8 (\Omega \cdot \text{cm})^{-1}$$

$$\text{p-type: } \sigma = e\mu_p p_o = (1.6 \times 10^{-19})(400)(2 \times 10^{16}) = 1.28 (\Omega \cdot \text{cm})^{-1}$$

$$\text{compensated: } \sigma = e\mu_n n_o = (1.6 \times 10^{-19})(1000)(3 \times 10^{16}) = 4.8 (\Omega \cdot \text{cm})^{-1}$$

$$\text{(c) } J = \sigma E \Rightarrow E = \frac{J}{\sigma}$$

$$\text{n-type: } E = \frac{120}{8.8} = 13.6 \text{ V/cm}$$

$$\text{p-type: } E = \frac{120}{1.28} = 93.75 \text{ V/cm}$$

$$\text{compensated: } E = \frac{120}{4.8} = 25 \text{ V/cm}$$

***5.43** In GaAs, the donor impurity concentration varies as $N_{d0} \exp(-x/L)$ for $0 \leq x \leq L$, where $L = 0.1 \mu\text{m}$ and $N_{d0} = 5 \times 10^{16} \text{ cm}^{-3}$. Assume $\mu_n = 6000 \text{ cm}^2/\text{V-s}$ and $T = 300 \text{ K}$. (a) Derive the expression for the electron diffusion current density versus distance over the given range of x . (b) Determine the induced electric field that generates a drift current density that compensates the diffusion current density.

(a) We have

$$J_{diff} = eD_n \frac{dn}{dx} = eD_n \frac{dN_d(x)}{dx} = \frac{eD_n}{(-L)} \cdot N_{d0} \exp\left(\frac{-x}{L}\right)$$

We have

$$D_n = \mu_n \left(\frac{kT}{e} \right) = (6000)(0.0259) \quad D_n = 155.4 \text{ cm}^2/\text{s}$$

Then

$$J_{diff} = \frac{-(1.6 \times 10^{-19})(155.4)(5 \times 10^{16})}{(0.1 \times 10^{-4})} \exp\left(\frac{-x}{L}\right) \quad J_{diff} = -1.243 \times 10^5 \exp\left(\frac{-x}{L}\right) \text{ A/cm}^2$$

(b)

$$0 = J_{drf} + J_{diff}$$

$$J_{drf} = e\mu_n nE = (1.6 \times 10^{-19})(6000)(5 \times 10^{16}) \left[\exp\left(\frac{-x}{L}\right) \right] E$$

$$J_{drf} = (48) \left[\exp\left(\frac{-x}{L}\right) \right] E$$

We have

$$J_{drf} = -J_{diff}$$

$$(48) \left[\exp\left(\frac{-x}{L}\right) \right] E = 1.243 \times 10^5 \exp\left(\frac{-x}{L}\right) \quad E = 2.59 \times 10^3 \text{ V/cm}$$

7.8 (a) Consider a uniformly doped silicon pn junction at $T = 300 \text{ K}$. At zero bias, 25 percent of the total space charge region is in the n-region. The built-in potential barrier is $V_{bi} = 0.710 \text{ V}$. Determine (i) N_a , (ii) N_d , (iii) x_n , (iv) x_p , and (v) $|E_{\max}|$. (b) Repeat part (a) for a GaAs pn junction with $V_{bi} = 1.180 \text{ V}$.

$$x_n = 0.25W = 0.25(x_n + x_p)$$

$$0.75x_n = 0.25x_p \Rightarrow \frac{x_p}{x_n} = 3$$

$$x_n N_d = x_p N_a \Rightarrow \frac{N_d}{N_a} = \frac{x_p}{x_n} = 3 \quad N_d = 3N_a$$

$$(a) \quad V_{bi} = (0.0259) \ln \left[\frac{N_a N_d}{(1.5 \times 10^{10})^2} \right]$$

$$0.710 = (0.0259) \ln \left[\frac{3N_a^2}{(1.5 \times 10^{10})^2} \right]$$

$$\text{or } 3N_a^2 = (1.5 \times 10^{10})^2 \exp \left(\frac{0.710}{0.0259} \right)$$

$$\text{which yields } N_a = 7.766 \times 10^{15} \text{ cm}^{-3} \quad N_d = 2.33 \times 10^{16} \text{ cm}^{-3}$$

$$x_n = \left\{ \frac{2 \epsilon_s V_{bi}}{e} \left(\frac{N_a}{N_d} \right) \left(\frac{1}{N_a + N_d} \right) \right\}^{1/2} = \left\{ \frac{2(11.7)(8.85 \times 10^{-14})(0.710)}{1.6 \times 10^{-19}} \times \left(\frac{1}{3} \right) \left[\frac{1}{4(7.766 \times 10^{15})} \right] \right\}^{1/2}$$

$$\Rightarrow x_n = 9.93 \times 10^{-6} \text{ cm}$$

$$\text{or } x_n = 0.0993 \mu \text{ m}$$

$$x_p = \left\{ \frac{2(11.7)(8.85 \times 10^{-14})(0.710)}{1.6 \times 10^{-19}} \times \left(\frac{3}{1} \right) \left[\frac{1}{4(7.766 \times 10^{15})} \right] \right\}^{1/2} = 2.979 \times 10^{-5} \text{ cm}$$

$$\text{or } x_p = 0.2979 \mu \text{ m}$$

$$|E_{\max}| = \frac{e N_d x_n}{\epsilon_s} = \frac{(1.6 \times 10^{-19})(2.33 \times 10^{16})(0.0993 \times 10^{-4})}{(11.7)(8.85 \times 10^{-14})} = 3.58 \times 10^4 \text{ V/cm}$$

(b) From part (a), we can write

$$3N_a^2 = (1.8 \times 10^6)^2 \exp \left(\frac{1.180}{0.0259} \right)$$

$$\text{which yields } N_a = 8.127 \times 10^{15} \text{ cm}^{-3}$$

$$N_d = 2.438 \times 10^{16} \text{ cm}^{-3}$$

$$x_n = \left\{ \frac{2(13.1)(8.85 \times 10^{-14})(1.180)}{1.6 \times 10^{-19}} \times \left(\frac{1}{3} \right) \left[\frac{1}{4(8.127 \times 10^{15})} \right] \right\}^{1/2}$$

$$= 1.324 \times 10^{-5} \text{ cm}$$

$$\text{or } x_n = 0.1324 \mu \text{ m}$$

$$x_p = \left\{ \frac{2(13.1)(8.85 \times 10^{-14})(1.180)}{1.6 \times 10^{-19}} \times \left(\frac{3}{1} \right) \left[\frac{1}{4(8.127 \times 10^{15})} \right] \right\}^{1/2} = 3.973 \times 10^{-5} \text{ cm}$$

or $x_p = 0.3973 \mu\text{m}$

$$|E_{\max}| = \frac{eN_d x_n}{\epsilon_s} = \frac{(1.6 \times 10^{-19})(2.438 \times 10^{16})(0.1324 \times 10^{-4})}{(13.1)(8.85 \times 10^{-14})} = 4.45 \times 10^4 \text{ V/cm}$$

- 7.19** A silicon n^+p junction is biased at $V_R = 5 \text{ V}$. (a) Determine the change in built-in potential barrier if the doping concentration in the p region increases by a factor of 3. (b) Determine the ratio of junction capacitance when the acceptor doping is $3N_a$ compared to that when the acceptor doping is N_a . (c) Why does the junction capacitance increase when the doping concentration increases?

$$\begin{aligned} \text{(a)} \quad & V_{bi}(3N_a) - V_{bi}(N_a) \\ &= V_t \ln \left[\frac{N_d(3N_a)}{n_i^2} \right] - V_t \ln \left[\frac{N_d N_a}{n_i^2} \right] = V_t \left\{ \ln(3) + \ln \left[\frac{N_d N_a}{n_i^2} \right] \right\} - V_t \ln \left[\frac{N_d N_a}{n_i^2} \right] = V_t \ln(3) = (0.0259) \ln(3) \\ &= 0.02845 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad & C' \cong \left\{ \frac{e \epsilon_s N_a}{2(V_{bi} + V_R)} \right\}^{1/2} \\ \text{So } & \frac{C'(3N_a)}{C'(N_a)} = \left\{ \frac{3N_a}{N_a} \right\}^{1/2} = \sqrt{3} = 1.732 \end{aligned}$$

- (c) For a larger doping, the space charge width narrows which results in a larger capacitance.

- 7.22** Consider a uniformly doped GaAs pn junction at $T = 300 \text{ K}$. The junction capacitance at zero bias is $C_j(0)$ and the junction capacitance with a 10-V reverse-biased voltage is $C_j(10)$. The ratio of the capacitances is

$$\frac{C_j(0)}{C_j(10)} = 3.13$$

Also under reverse bias, the space charge width into the p region is 0.2 of the total space charge width. Determine (a) V_{bi} and (b) N_a, N_d .

- (a) We have

$$\frac{C'_j(0)}{C'_j(10)} = \frac{\left[\frac{\epsilon_s N_a N_d}{2(V_{bi})(N_a + N_d)} \right]^{1/2}}{\left[\frac{\epsilon_s N_a N_d}{2(V_{bi} + V_R)(N_a + N_d)} \right]^{1/2}} \quad \frac{C'_j(0)}{C'_j(10)} = 3.13 = \left(\frac{V_{bi} + V_R}{V_{bi}} \right)^{1/2}$$

For $V_R = 10 \text{ V}$, we find

$$(3.13)^2 V_{bi} = V_{bi} + 10 \quad V_{bi} = 1.137 \text{ V}$$

- (b) $x_p = 0.2W = 0.2(x_p + x_n)$

$$\frac{x_p}{x_n} = 0.25 = \frac{N_d}{N_a}$$

$$V_{bi} = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

$$1.137 = (0.0259) \ln \left[\frac{0.25 N_a^2}{(1.8 \times 10^6)^2} \right]$$

$$N_a = \frac{1.8 \times 10^6}{\sqrt{0.25}} \exp \left[\frac{1.137}{2(0.0259)} \right]$$

$$N_a = 1.23 \times 10^{16} \text{ cm}^{-3}$$

$$N_d = 3.07 \times 10^{15} \text{ cm}^{-3}$$

- 8.5** Consider a GaAs pn junction with doping concentrations $N_a = 5 \times 10^{16} \text{ cm}^{-3}$ and $N_d = 10^{16} \text{ cm}^{-3}$. The junction cross-sectional area is $A = 10^{-3} \text{ cm}^2$ and the applied forward-bias voltage is $V_a = 1.10 \text{ V}$. Calculate the (a) minority electron diffusion current at the edge of the space charge region, (b) minority hole diffusion current at the edge of the space charge region, and (c) total current in the pn junction diode.

$$\begin{aligned} \text{(a)} \quad J_n(-x_p) &= \frac{eD_n n_{po}}{L_n} \exp \left(\frac{V_a}{V_t} \right) = \frac{en_i^2}{N_a} \sqrt{\frac{D_n}{\tau_{no}}} \cdot \exp \left(\frac{V_a}{V_t} \right) = \frac{(1.6 \times 10^{-19})(1.8 \times 10^6)^2}{5 \times 10^{16}} \sqrt{\frac{205}{5 \times 10^{-8}}} \times \exp \left(\frac{1.10}{0.0259} \right) \\ &= 1.849 \text{ A/cm}^2 \end{aligned}$$

$$I_n = AJ_n(-x_p) = (10^{-3})(1.849) \text{ A}$$

$$\text{or } I_n = 1.85 \text{ mA}$$

$$\begin{aligned} \text{(b)} \quad J_p(x_n) &= \frac{eD_p p_{no}}{L_p} \exp \left(\frac{V_a}{V_t} \right) = \frac{en_i^2}{N_d} \sqrt{\frac{D_p}{\tau_{p0}}} \cdot \exp \left(\frac{V_a}{V_t} \right) = \frac{(1.6 \times 10^{-19})(1.8 \times 10^6)^2}{10^{16}} \sqrt{\frac{9.80}{10^{-8}}} \times \exp \left(\frac{1.10}{0.0259} \right) \\ &= 4.521 \text{ A/cm}^2 \end{aligned}$$

$$I_p = AJ_p(x_n) = (10^{-3})(4.521) \text{ A}$$

$$\text{or } I_p = 4.52 \text{ mA}$$

$$\text{(c)} \quad I = I_n + I_p = 1.85 + 4.52 = 6.37 \text{ mA}$$

- 8.8** A one-sided p⁺n silicon diode has doping concentrations of $N_a = 5 \times 10^{17} \text{ cm}^{-3}$ and $N_d = 8 \times 10^{15} \text{ cm}^{-3}$. The minority carrier lifetimes are $\tau_{n0} = 10^{-7} \text{ s}$ and $\tau_{p0} = 8 \times 10^{-8} \text{ s}$. The cross-sectional area is $A = 2 \times 10^{-4} \text{ cm}^2$. Calculate the (a) reverse-biased saturation current, and (b) the forward-bias current at (i) $V_a = 0.45 \text{ V}$, (ii) $V_a = 0.55 \text{ V}$, and (iii) $V_a = 0.65 \text{ V}$.

$$(a) \quad J_s = en_i^2 \left[\frac{1}{N_a} \sqrt{\frac{D_n}{\tau_{no}}} + \frac{1}{N_d} \sqrt{\frac{D_p}{\tau_{p0}}} \right] = (1.6 \times 10^{-19}) (1.5 \times 10^{10})^2 \times \left[\frac{1}{5 \times 10^{17}} \sqrt{\frac{25}{10^{-7}}} + \frac{1}{8 \times 10^{15}} \sqrt{\frac{10}{8 \times 10^{-8}}} \right]$$

$$J_s = 5.145 \times 10^{-11} \text{ A/cm}^2$$

$$I_s = AJ_s = (2 \times 10^{-4}) (5.145 \times 10^{-11}) = 1.029 \times 10^{-14} \text{ A}$$

$$(b) \quad I = I_s \exp\left(\frac{V_a}{V_t}\right)$$

$$(i) \quad I = (1.029 \times 10^{-14}) \exp\left(\frac{0.45}{0.0259}\right) = 3.61 \times 10^{-7} \text{ A}$$

$$(ii) \quad I = (1.029 \times 10^{-14}) \exp\left(\frac{0.55}{0.0259}\right) = 1.72 \times 10^{-5} \text{ A}$$

$$(iii) \quad I = (1.029 \times 10^{-14}) \exp\left(\frac{0.65}{0.0259}\right) = 8.16 \times 10^{-4} \text{ A}$$